

1. Causal inference: compare potential outcomes, not observed histories

Effects of a cause

Ask what a **well-defined, manipulable treatment** does to an outcome—not merely what caused an observed event.

Potential outcomes

For unit i and binary treatment $T_i \in \{0, 1\}$:

$Y_i(1)$ = outcome under treatment,

$Y_i(0)$ = outcome under control.

The individual causal effect is the comparison

$$\tau_i = Y_i(1) - Y_i(0).$$

The assignment mechanism reveals exactly one outcome:

$$Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0).$$

Treatment selects which potential outcome becomes *factual*.

The fundamental problem

Unit	T_i	$Y_i(1)$	$Y_i(0)$
1	1	1	?
2	1	1	?
3	0	?	1
4	0	?	0

We never observe both the factual and counterfactual outcome for the same unit at the same time.

- ▶ Individual effects are therefore not directly observable.
- ▶ Causal inference is a **missing-data problem**: a design must supply credible stand-ins for the unobserved outcomes.
- ▶ The target is usually an **average** causal effect over a population.

2. Correlation can miss causation: assignment determines the comparison

Why the naïve difference can be wrong

The observed treatment–control contrast is

$$\Delta_{\text{obs}} = \underbrace{E[Y(1) - Y(0) \mid T = 1]}_{\text{ATT}} + \underbrace{E[Y(0) \mid T = 1] - E[Y(0) \mid T = 0]}_{\text{selection bias}}.$$

If people select treatment, or a programme targets those expected to benefit, treated and control units are not exchangeable.

Correlation is neither necessary nor sufficient

- ▶ A common cause can generate association with no causal effect.
- ▶ Optimising behaviour can hide a real effect: the sailor moves the rudder *because of* the wind, keeping the boat straight.
- ▶ Outcomes may predict treatment *after* treatment; the problem is when the full schedule $(Y(1), Y(0))$ predicts assignment.

When group comparisons recover truth

We need treatment assignment to be unrelated to potential outcomes:

$$(Y(1), Y(0)) \perp\!\!\!\perp T.$$

Then treated and control units represent the same underlying response distributions:

$$\begin{aligned} E[Y(1) \mid T = 1] &= E[Y(1)], \\ E[Y(0) \mid T = 0] &= E[Y(0)]. \end{aligned}$$

Hence

$$\hat{\tau}_{\text{DiM}} = \bar{Y}_{T=1} - \bar{Y}_{T=0}$$

estimates the average treatment effect.

Random assignment promotes this independence by making treatment status a chance event rather than a choice based on likely outcomes.

Design lesson: learn how treatment was assigned before interpreting an association causally.

3. Estimands and assumptions: state exactly which causal effect you mean

From individual effects to averages

$$\tau_i = Y_i(1) - Y_i(0), \quad \text{ATE} = E[Y(1) - Y(0)].$$
$$\text{ATT} = E[Y(1) - Y(0) \mid T = 1].$$

ATE describes the target population; ATT describes those who actually received treatment. They coincide only under additional conditions.

Statistical language

- ▶ **Estimand:** the target quantity (for example, ATE).
- ▶ **Estimator:** a rule computed from observed data (for example, difference in means).
- ▶ **Estimate:** the numerical value returned by the estimator.

SUTVA and treatment definition

A causal comparison requires:

- ▶ a well-defined treatment and control;
- ▶ **no hidden versions** with different effects;
- ▶ **no interference:** unit i 's outcome depends only on i 's treatment;
- ▶ assignment affects outcomes only through the treatment being studied.

Do not confuse three claims

$$\begin{aligned} \text{constant effect: } & \tau_i = \tau \quad \forall i, \\ \text{zero average effect: } & E[\tau_i] = 0, \\ \text{sharp null: } & \tau_i = 0 \quad \forall i. \end{aligned}$$

The sharp null is much stronger: it rules out an effect for *every* unit, not just on average.

4. Experiments identify effects—but assignment and receipt may differ

Why experiments work

Random assignment makes the missing potential outcomes recoverable in expectation:

$$(Y(1), Y(0)) \perp\!\!\!\perp T \\ \Rightarrow \text{ATE} = E[Y \mid T = 1] - E[Y \mid T = 0].$$

Randomisation identifies an effect through the **design**; the difference in means is the corresponding calculation.

Assignment is not always receipt

Let T_i denote assigned treatment and D_i treatment received. Noncompliance creates four latent types:

Type	$D_i(1)$	$D_i(0)$
Complier	1	0
Always-taker	1	1
Never-taker	0	0
Defier	0	1

Preserve the random comparison

The intent-to-treat effect compares outcomes by assignment:

$$\text{ITT} = E[Y \mid T = 1] - E[Y \mid T = 0].$$

It answers the policy-relevant question: what is the effect of *offering or assigning* treatment?

The as-treated contrast,

$$E[Y \mid D = 1] - E[Y \mid D = 0],$$

can reintroduce selection because receipt is partly chosen. It is not automatically causal.

Computing toolkit in R

- ▶ Build vectors with `c()`, `:`, `seq()`, and `rep()`.
- ▶ Subset with `[ ]`; combine logical tests to define treatment and control groups.
- ▶ Use named arguments and small functions to make estimators explicit and reproducible.

Workflow: define the estimand → understand assignment → compute the design-based contrast.